Lecture 11.
How to Train Large Models

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21.04.2025

Training Steps

- Pretraining
  - Model trained as the next word predictor
  - Model encodes knowledge about the world
- Supervised Fine-Tuning
  - Adapts the model for specific tasks, such as question answering, chatting, or coding
  - (in particular Instruction Tuning) makes model better at following instructions
- Preference Tuning
  - Makes model outputs align better with human preferences

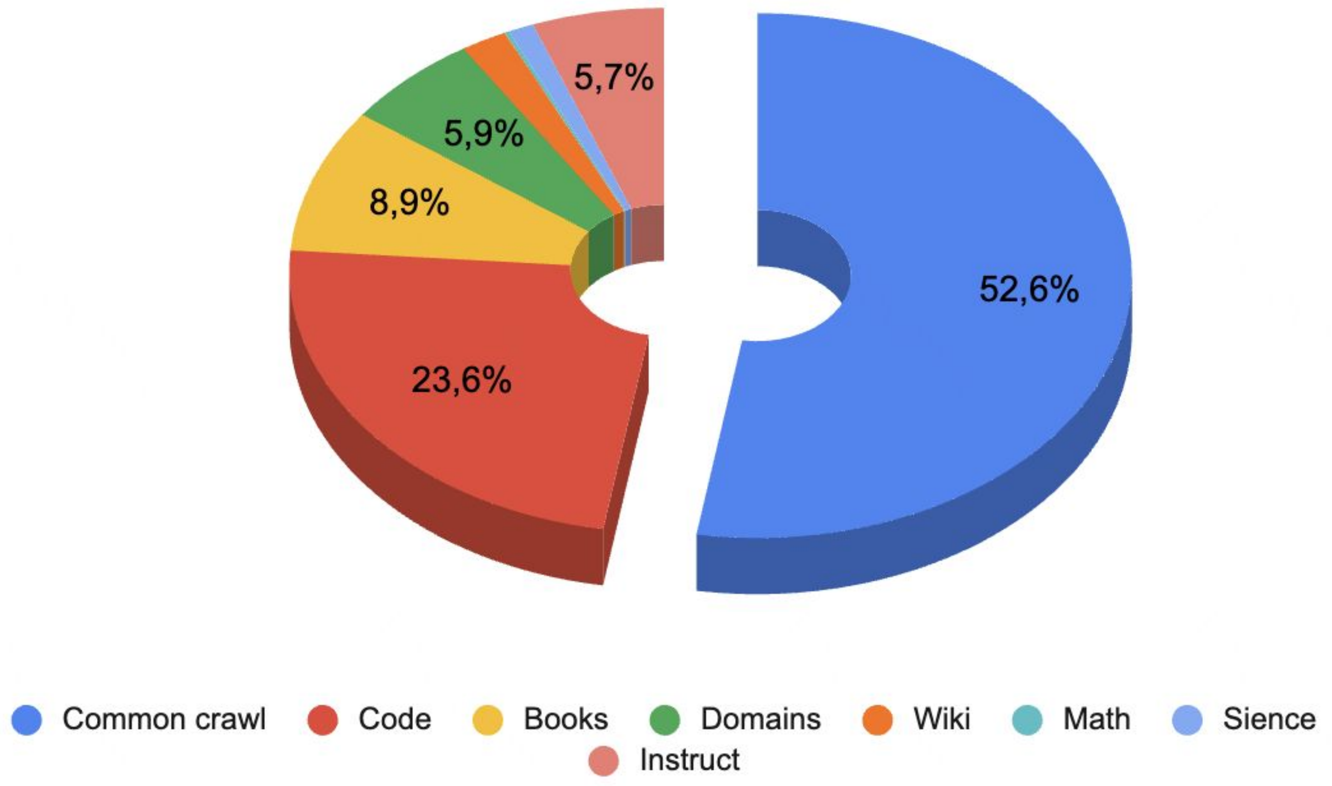
Pretraining

- Data Collection
- Data Preprocessing
- Architecture Choice

Data Collection

- General Text Data
  - Web Pages (CommonCrawl)
  - Conversation Text (Reddit)
  - Books (Google Books)
- Specialized Text Data
  - Multilingual Text (bloom library)
  - Scientific Text (arXiv papers, scientific textbooks, math websites)
  - Code (GitHub)

Data Collection



T-lite and T-pro: training report

Data Preprocessing

Good Data

- Unique and Distinct
- Accurate and Error-Free
- Consistent
- Validated
- Safe and Secure

Bad Data

- Duplicated
- Typos and Spelling Errors
- Inconsistent
- Hallucinations
- Toxic

Cleaning Pipeline

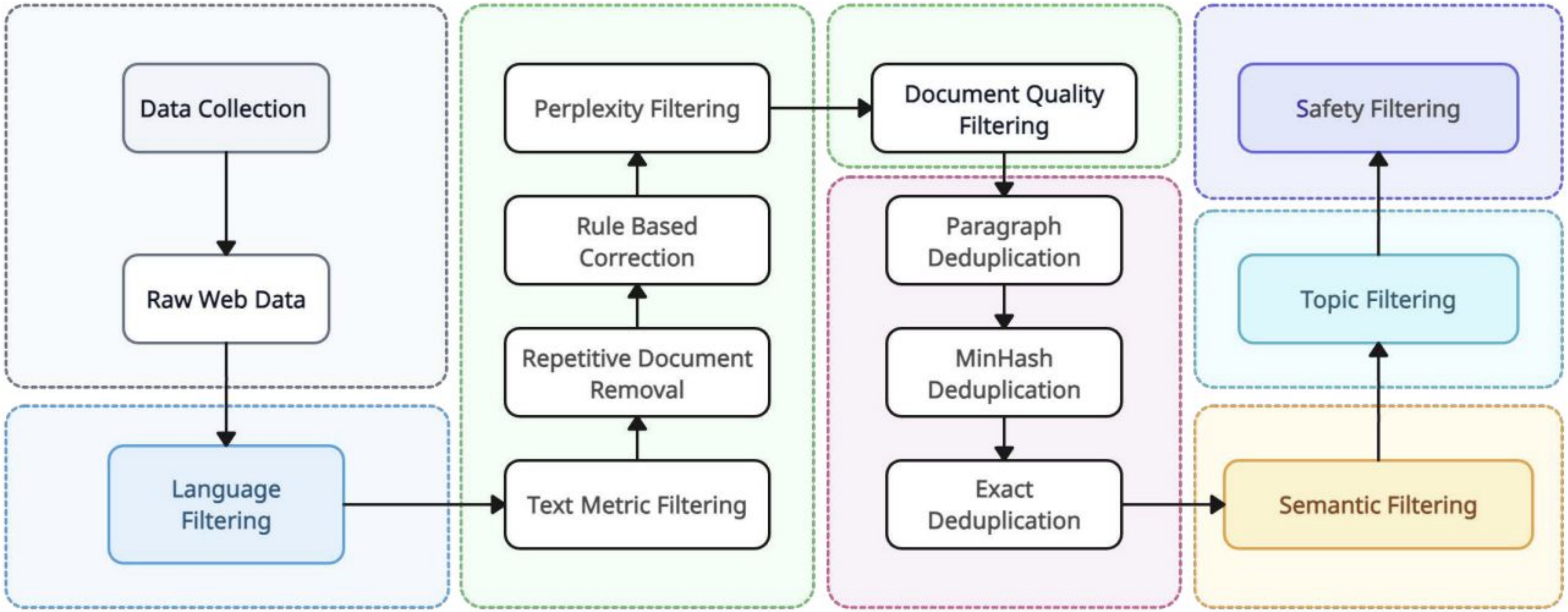


Figure 1: Yi's pretraining data cleaning pipeline.

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Filtering Methods

- Noise Removal
- Language Filtering
- Text Quality
- Ethical Filtering
- Duplicate Removal

Filtering Methods

- **Noise Removal**
  - Remove HTML tags, JavaScript code, CSS, and other technical elements
  - Remove repeated characters, random character sequences
  - Remove documents that are too short or too long
- Language Filtering
- Text Quality
- Ethical Filtering
- Duplicate Removal

Filtering Methods

- Noise Removal
- **Language Filtering**
  - Models such as FastText or langdetect are used to determine the language of the text
  - Only texts in the target language are retained
- Text Quality
- Ethical Filtering
- Duplicate Removal

Filtering Methods

- Noise Removal
- Language Filtering
- **Text Quality**
  - Evaluate "perplexity" using a small language model. High perplexity may indicate low-quality text
  - Remove texts with a large number of grammatical errors or unnatural constructions
- Ethical Filtering
- Duplicate Removal

Filtering Methods

- Noise Removal
- Language Filtering
- Text Quality
- **Ethical Filtering**
  - Remove texts containing profanity, insults, disinformation, or malicious content.
  - Pre-trained classifiers are used to detect toxic or biased texts
- Duplicate Removal

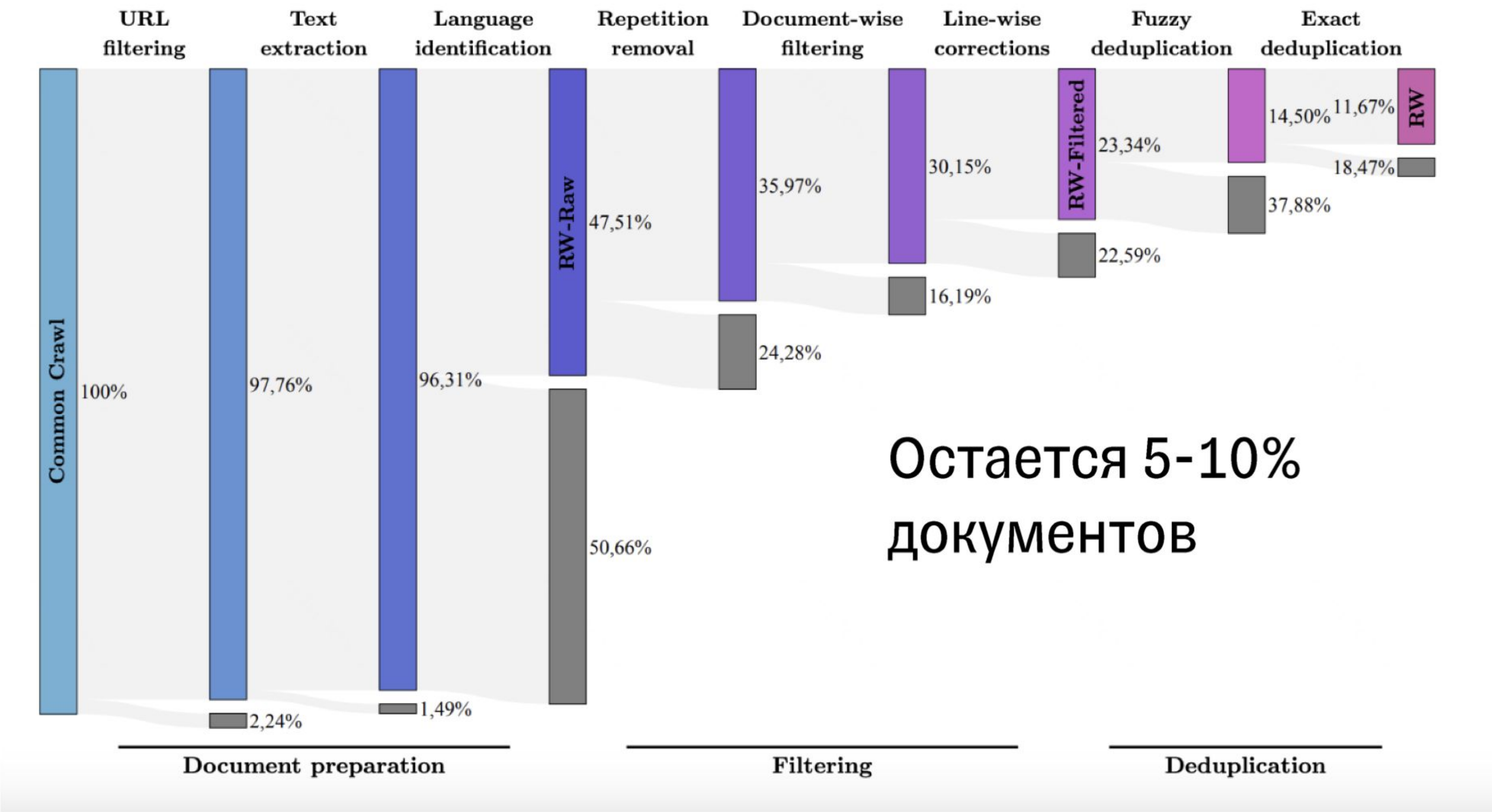
Filtering Methods

- Noise Removal
- Language Filtering
- Text Quality
- Ethical Filtering
- **Duplicate Removal**
    - Check for exact duplicates (completely matching texts)
    - Check for partial duplicates (texts with a high degree of similarity). Algorithms such as MinHash or SimHash can be used for this

Data Deduplication

- Exact Duplicates
  - Hashing texts (e.g., using SHA-256) and comparing hashes
- Partial Duplicates
  - Using algorithms such as MinHash or SimHash to detect similar texts
  - Clustering texts based on their embeddings (e.g., using BERT or Sentence-BERT)
- Cross-Document Deduplication
  - Removing texts that appear in multiple sources (e.g., identical articles on different websites)

Cleaning Pipeline



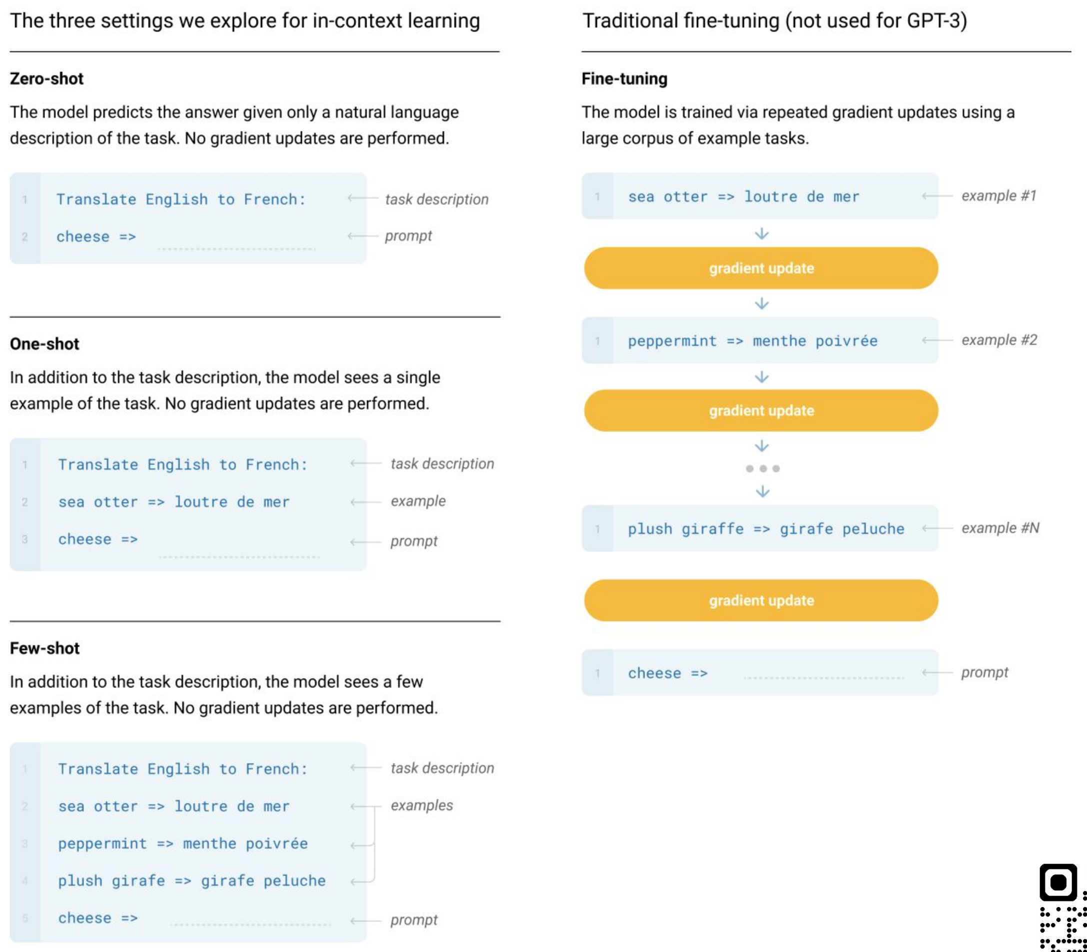
[T-Bank NLP Meetup]

Architecture Choice

- **Encoder-Only** Models
  - Example: BERT-based models
  - Pretraining Approach: Masked Language Modelling (MLM)
- **Decoder-Only** Models
  - Example: GPT, LLaMA
  - Pretraining Approach: Next Token Prediction
- **Encoder-Decoder** Models
  - Example: T5, BART
  - Pretraining: Task-dependent

Why Decoder-Only Models?

- **Encoder-Decoder** model achieve its maximum potential by multitask fine-tuning on labeled data
- **Casual Decoder** only requires self-supervised learning on a large-scale corpus



[2005.14165](https://arxiv.org/abs/2005.14165)

Scaling Laws

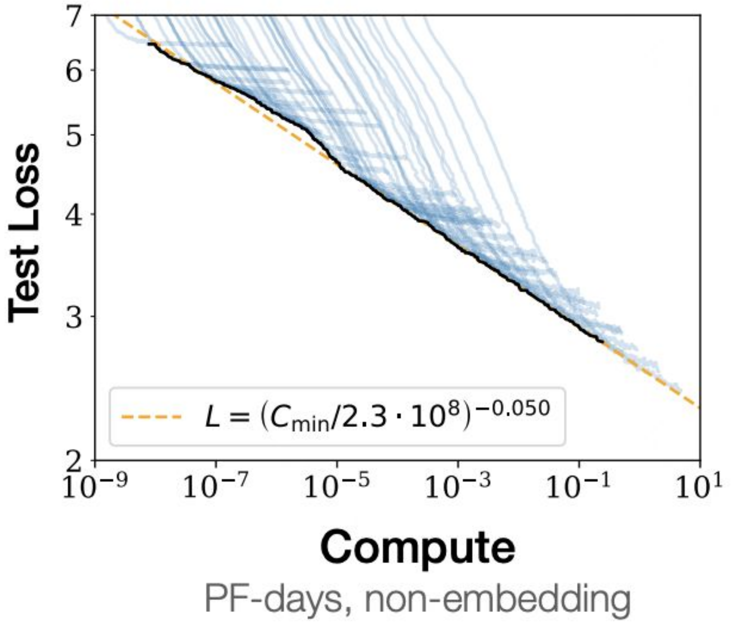


Figure 1 Language modeling performance improves smoothly as we increase the model size, datasetset size, and amount of compute[2] used for training. For optimal performance all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two.

[2001.08361](2001.08361)

Chinchilla scaling

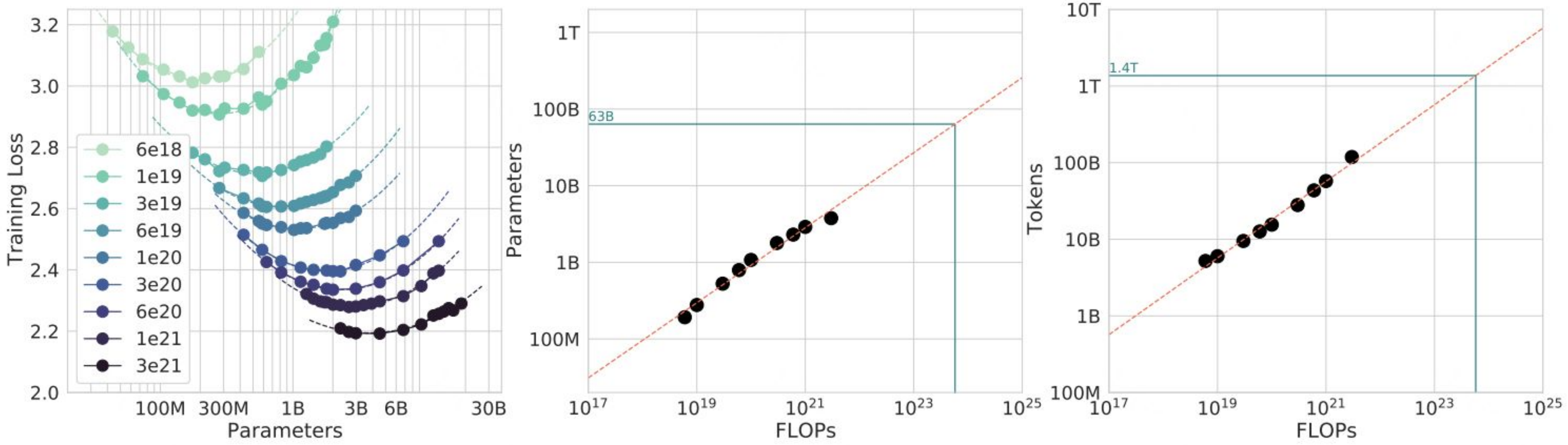


Figure 3 | **IsoFLOP curves.** For various model sizes, we choose the number of training tokens such that the final FLOPs is a constant. The cosine cycle length is set to match the target FLOP count. We find a clear valley in loss, meaning that for a given FLOP budget there is an optimal model to train (**left**). Using the location of these valleys, we project optimal model size and number of tokens for larger models (**center** and **right**). In green, we show the estimated number of parameters and tokens for an *optimal* model trained with the compute budget of *Gopher*.

[2203.15556](2203.15556)

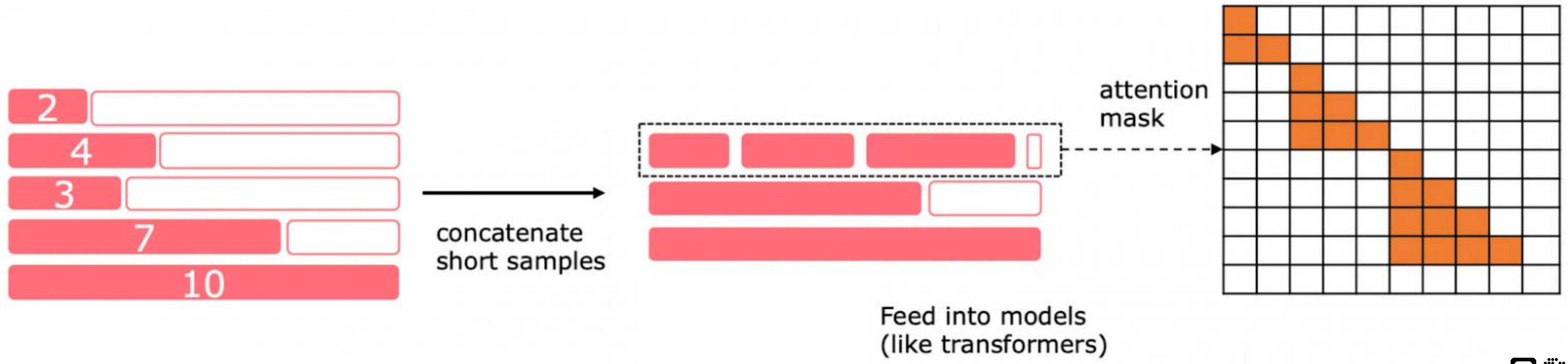
Training without Padding

Disadvantages of Padding

- **Wasted Computation.** The model performs calculations even for "empty" tokens, reducing efficiency
- **Sequence Length Limitation.** If padding is used, the maximum sequence length is limited by the batch size, which can be problematic for long texts

Alternative: Text Concatenation

1. Text Concatenation
2. Splitting into Batches
3. Masking



[T-lite and T-pro: training report]

Alternative: Text Concatenation

1. **Text Concatenation**
   ○ All texts from the corpus are combined into one long sequence, separated by special tokens (e.g., [EOS])
2. Splitting into Batches
3. Masking

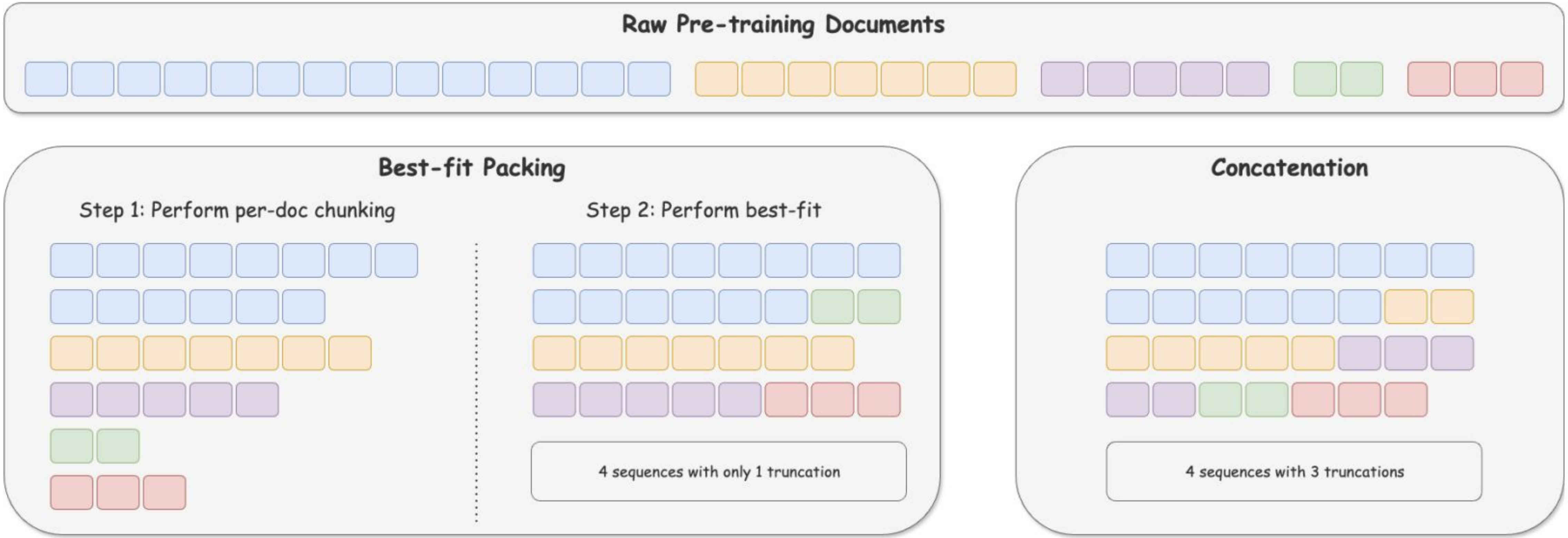
Alternative: Text Concatenation

1. Text Concatenation
2. **Splitting into Batches**
    - The long sequence is divided into chunks of fixed length
    - Each chunk becomes a separate batch
3. Masking

Alternative: Text Concatenation

1. Text Concatenation
2. Splitting into Batches
3. **Masking**
    - A mask is created for each batch to indicate where one text ends and another begins
    - This allows the model to correctly handle text boundaries

Alternative: Text Concatenation



[2404.10830](2404.10830)

References

T-lite and T-pro: training report

- https://habr.com/ru/companies/tbank/articles/890236/

Yi: Open Foundation Models by 01.AI

- https://arxiv.org/pdf/2403.04652

Deduplicating Training Data Makes Language Models Better

- https://arxiv.org/pdf/2107.06499

Language Models are Few-Shot Learners

- https://arxiv.org/pdf/2005.14165

Fewer Truncations Improve Language Modeling

- https://arxiv.org/pdf/2404.10830

Self-Study Materials

Improving Language Understanding by Generative Pre-Training

- https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf

To Code, or Not To Code? Exploring Impact of Code in Pre-training

- https://arxiv.org/pdf/2408.10914v1

Scaling Laws for Neural Language Models

- https://arxiv.org/pdf/2001.08361

Training Compute-Optimal Large Language Models

- https://arxiv.org/pdf/2203.15556